SOFTWARE ENGINEERING ACADEMIC PROJECT — MID-SEMESTER EVALUATION

Department of Computer Science & Engineering

Database Design & Data Model

Project: AutoTestHub (Web Automation & Testing Platform)

ENTITY RELATIONSHIP • SCHEMA SPECIFICATION • MONGOOSE ODM

DATABASE ARCHITECTURE METADATA

Database Engine: MongoDB (v7.0+)

Object Data Modeling: Mongoose ODM (v9.11)

Design Paradigm: Document Store with Normalized References

Connection URI: `mongodb://127.0.0.1:27017/autotesthub`

PROJECT DEVELOPMENT TEAM

Shristi Neekhra (Integration Lead)

Shifa Siddiqui (Database & Backend Lead)

Shruti Gupta (Selenium Engine Architect)

Ishree (Frontend Specialist)

Academic Guide: [Supervisor Name]

1. Data Modeling Strategy & Paradigm

AutoTestHub utilizes MongoDB, a distributed NoSQL document database, coupled with Mongoose for schema enforcement. The choice of document storage is driven by:

- **Polymorphic Action Structures:** Individual test actions vary significantly in structure (e.g. `OPEN_URL` requires only a target URL, whereas `TYPE` requires `locatorType`, `locator`, and `value`). MongoDB's flexible BSON document model stores these polymorphic action arrays natively without complex relational table joins.
- **Temporal Test Audit History:** Test execution outcomes are append-mostly time-series events. A normalized 1-to-Many relationship between `TestCase` and `TestResult` allows efficient historical trending and regression analysis.

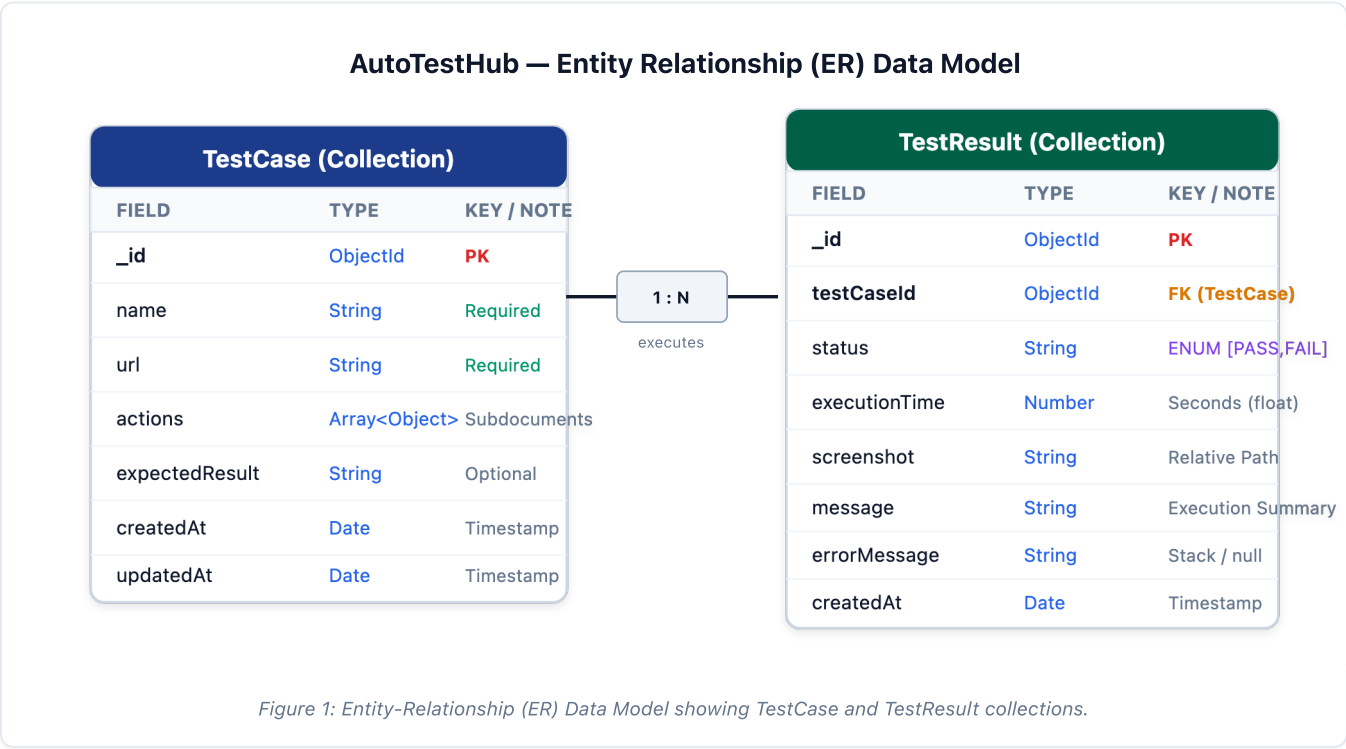


Figure 1: Entity-Relationship (ER) Data Model showing TestCase and TestResult collections.

1.1 Entity Relationship Cardinality

The system maintains a **1 : N (One-to-Many)** relationship between `TestCase` and `TestResult`:

- One `TestCase` definition may be executed multiple times over the project lifecycle, producing zero, one, or many corresponding `TestResult` records.
- Each `TestResult` document maintains a mandatory foreign key reference (`testCaseId`) pointing to the parent `TestCase._id`.

2. Collection Specifications & Data Dictionary

2.1 `TestCase` Collection Schema

Field Name	BSON Type	Constraints	Default	Description
`_id`	ObjectId	Primary Key	Auto-generated	Unique document identifier.
`name`	String	Required, Trimmed	None	Human-readable descriptive test name.
`url`	String	Required	None	Base target URL for test navigation.
`actions`	Array<Object>	Array of subdocuments	`[]`	Ordered list of browser action descriptors.
`expectedResult`	String	Optional	`""`	High-level expected outcome summary.
`createdAt`	Date	Mongoose Timestamp	Current Date	Creation timestamp.
`updatedAt`	Date	Mongoose Timestamp	Current Date	Last modification timestamp.

2.2 `TestResult` Collection Schema

Field Name	BSON Type	Constraints	Default	Description
`_id`	ObjectId	Primary Key	Auto-generated	Unique execution record identifier.
`testCaseId`	ObjectId	Required, Ref: 'TestCase'	None	Foreign key reference to parent test case.
`status`	String	Enum: ['PASSED', 'FAILED']	None	Final execution verdict.
`executionTime`	Number (Float)	Non-negative	0	Duration of browser run in seconds.
`screenshot`	String	Optional	null	Relative filesystem path to captured PNG.
`message`	String	Optional	""	Execution summary message.
`errorMessage`	String	Optional	null	Exception stack trace or failure detail.
`createdAt`	Date	Mongoose Timestamp	Current Date	Execution completion timestamp.

2.3 Sample BSON Document Records

TestCase Document:

```
{
  "_id": ObjectId("6ad5f308777225dd7c9950a4"),
  "name": "Full Integration Pipeline Test",
  "url": "https://example.com",
  "actions": [
    { "type": "OPEN_URL" },
    { "type": "ASSERT_TITLE", "expected": "Example Domain" }
  ],
  "expectedResult": "Page title matches 'Example Domain'",
  "createdAt": ISODate("2026-10-19T10:38:00.000Z"),
  "updatedAt": ISODate("2026-10-19T10:38:00.000Z")
}
```

TestResult Document:


```
{
  "_id": ObjectId("6ad5f308777225dd7c9950b7"),
  "testCaseId": ObjectId("6ad5f308777225dd7c9950a4"),
  "status": "PASSED",
  "executionTime": 1.48,
  "screenshot": "screenshots/6ad5f308777225dd7c9950a4_20261019_160809_9fb515.png",
  "message": "Page title matched 'Example Domain'",
  "errorMessage": null,
  "createdAt": ISODate("2026-10-19T10:38:09.000Z")
}
```